Urban Policies to Combat Climate Change: Assessing the Impact of Congestion Pricing and Low-Emission Zones using Machine Learning Techniques

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July 30, 2026

Course: *Machine Learning in Econometrics*
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Executive Summary

Abstract content goes here. This section should provide a concise summary of the research question, methodology, key findings, and implications of the study. It should be clear and informative, allowing readers to quickly understand the purpose and results of the paper.

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Replication files are available in the GitHub repository for this project.

1 Introduction

2 Empirical Strategy and Identification

- Target parameters: ATE, ATT, GATT (possible differences in causal forest and aipw lasso estimation) (show math and explain shortly)

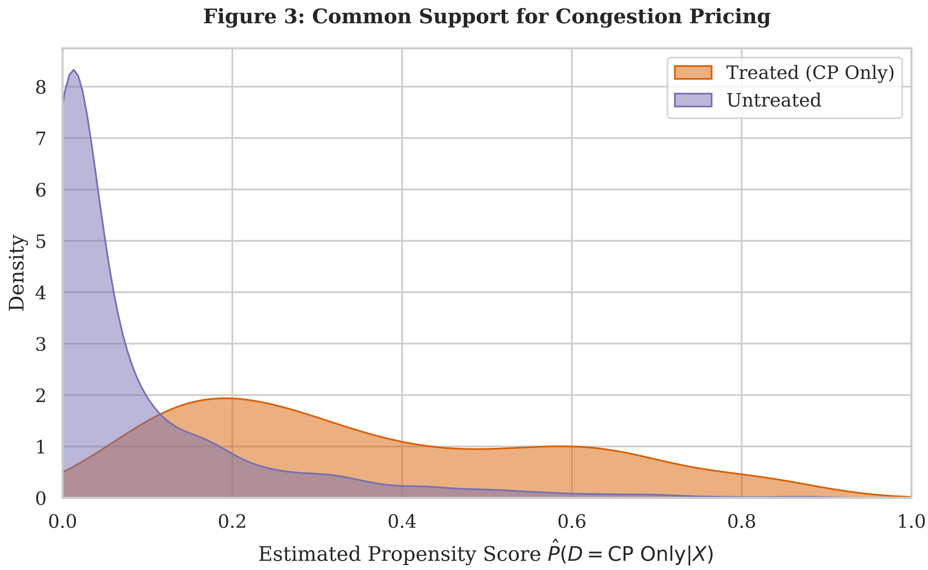

Figure 1: Common Support for Congestion Pricing (AIPW)

3 Proposed Estimator

- Use AIPW and Causal Forest instead of IPW or Reg. adj. - Explain how they work and differences

4 Theoretical Properties

- Proof of Neymon Orthogonality (is it the same for both?) - Not yet sure about other properties

5 Data Preparation

- no missing values or absurd numbers - check for skewed distributions (plot in appendix) - log transform variables where it makes economic sense and where it solves skewness - Time invariant variables and creation of Mundlak device in order to account for panel structure (national climate pact removed because not changing)

6 K-means clustering

- determination of two clusters necessary to make heterogeneity analysis with many covariates viable - optimal k = 2 (interpretability and silhouette analysis) - Cluster 1: urban metropolises, Cluster 2: sprawling regional hubs - describe them based on the cluster differences plot

Table 1: Distribution of Climate Policy Treatments by Urban Typology

Urban Typology	Never Treated	CP Only	LEZ Only	Synergy (CP $\times$ LEZ)	Total
Dense Metropolises (1)	11	27	16	25	79
Regional Hubs (0)	62	13	26	3	104
Total	**73**	**40**	**42**	**28**	**183**

Notes: Table displays the count of unique cities. A city is classified as "Ever Treated" if Congestion Pricing was active in any year during the sample period.

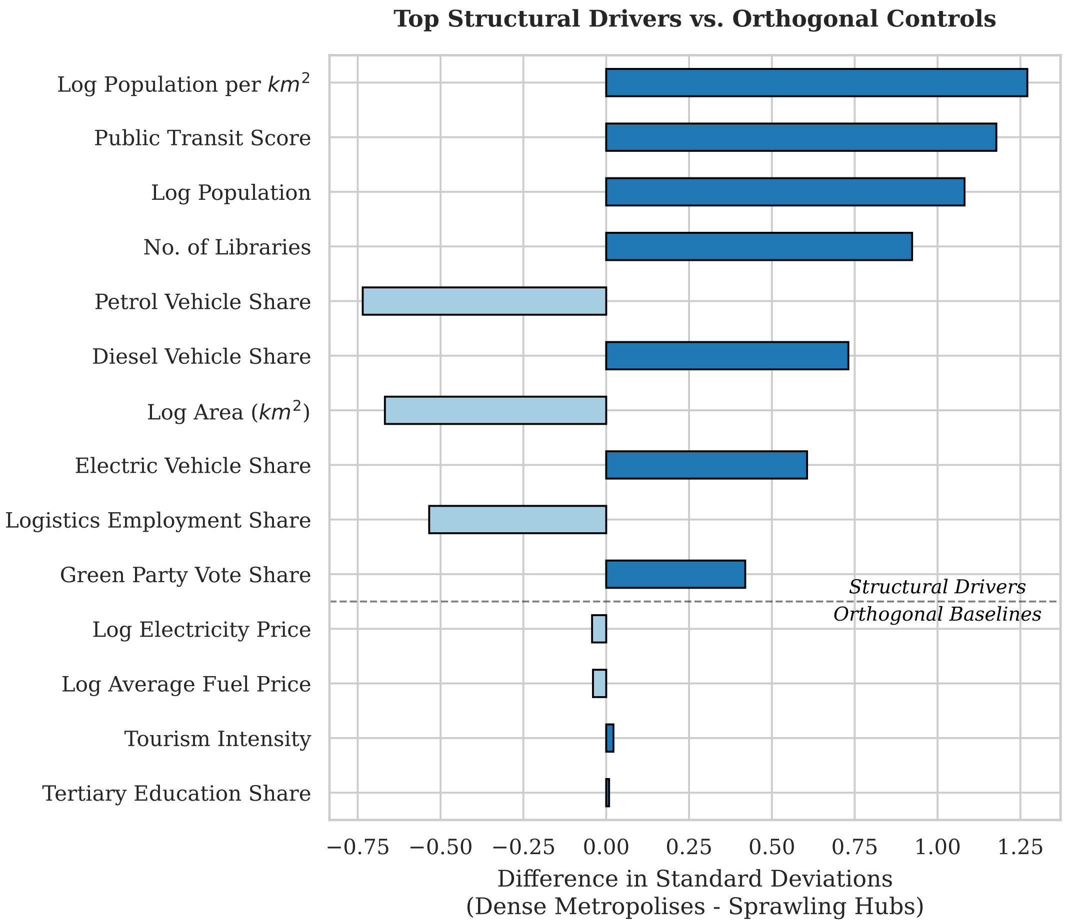

Figure 2: Differences between the two clusters.

Table 2: Double Machine Learning Estimates of Urban Climate Policies (Categorical Regime)

Policy	(1) AIPW	(2) Causal Forest
Panel A: Average Treatment Effect on the Treated (ATT)		
Congestion Pricing (CP)	-0.3047 ($p = 0.131$)	-0.3757^{***} ($p = 0.000$)
Low Emission Zone (LEZ)	-0.1121 ($p = 0.141$)	-0.1638^{**} ($p = 0.040$)
Synergy (CP × LEZ)	-1.3543^{***} ($p = 0.000$)	-1.2471^{***} ($p = 0.000$)
Panel B: Group ATT (Heterogeneity by City Type)		
Cluster 1: Dense Metropolis		
Congestion Pricing (CP)	-0.3141 ($p = 0.260$)	-0.3799^{***} ($p = 0.004$)
Low Emission Zone (LEZ)	-0.1457 ($p = 0.191$)	-0.2331^{**} ($p = 0.037$)
Synergy (CP × LEZ)	-1.4209^{***} ($p = 0.000$)	-1.3163^{***} ($p = 0.000$)
Cluster 0: Sprawling Cities		
Congestion Pricing (CP)	-0.2775^{***} ($p = 0.002$)	-0.3634^{***} ($p = 0.001$)
Low Emission Zone (LEZ)	-0.0774 ($p = 0.302$)	-0.0922 ($p = 0.353$)
Synergy (CP × LEZ)	-0.7277^{**} ($p = 0.041$)	-0.5955^{**} ($p = 0.019$)

Notes: p-values in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. 'Failed' estimation bounds denote overlap violations.

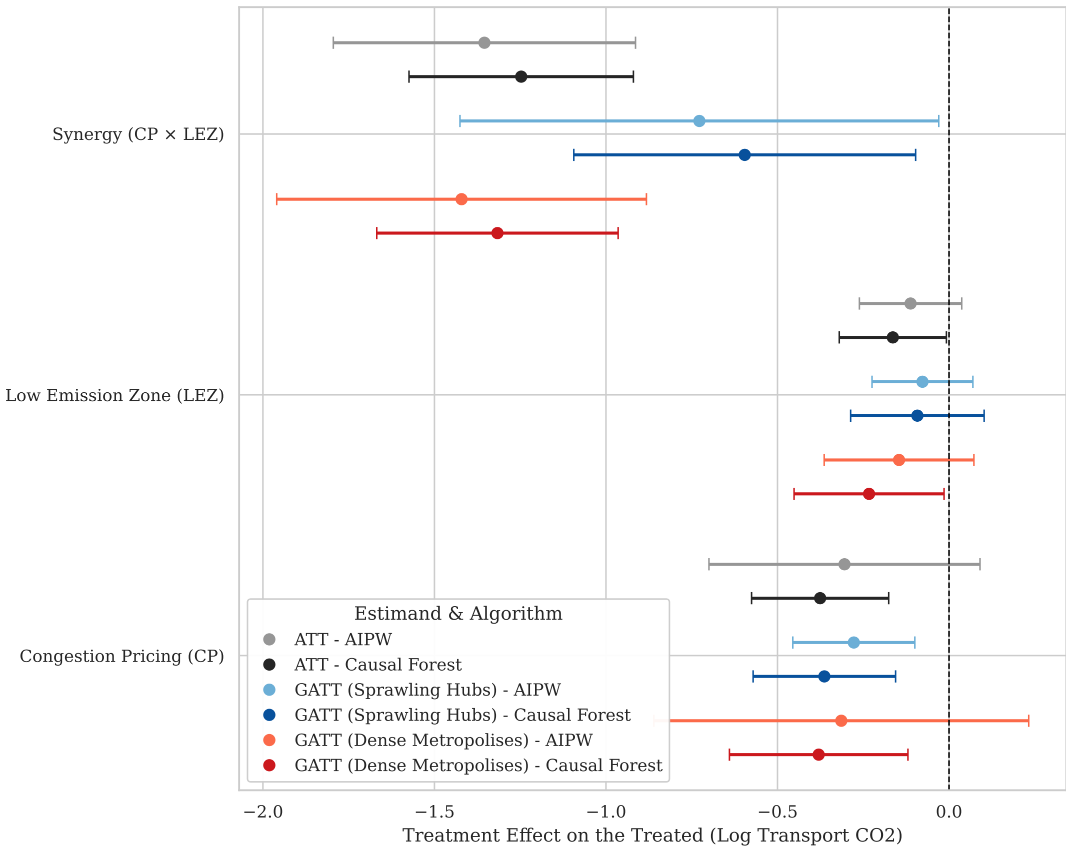

Figure 3: Policy Efficacy by Algorithm (95% CI)

7 Results

8 Sensitivity Analysis

Table 3: Propensity Score Distribution and Common Support (Congestion Pricing)

Statistic	Treated (CP Only)	Untreated
Observations (N)	380	2731
Mean	0.359	0.085
Std. Dev.	0.224	0.133
Minimum	0.004	0.000
Median	0.301	0.026
Maximum	0.872	0.885

Notes: The common support region is strictly defined as $[0.0043, 0.8722]$. Within this bounded interval, the analysis retains 380 treated units (100.0%) and 2092 untreated units (76.6%).

Table 4: Falsification Tests: Estimated Effects on Placebo Outcomes (Global ATT)

Decoy Outcome	Congestion Pricing (CP)	Low Emission Zone (LEZ)	Synergy (CP × LEZ)
Panel A: AIPW (Lasso)			
No. of Libraries	-0.0370 ($p = 0.184$)	0.0606 ($p = 0.216$)	0.1269 ($p = 0.571$)
Streetlight Density	0.8913 ($p = 0.470$)	-0.8973 ($p = 0.215$)	-0.4583 ($p = 0.861$)
No. of Fountains	-0.3113 ($p = 0.383$)	-0.1112 ($p = 0.652$)	-0.3178 ($p = 0.470$)
No. of benches pc	0.0205 ($p = 0.830$)	-0.4880^{***} ($p = 0.000$)	-0.1854 ($p = 0.234$)
Panel B: Causal Forest			
No. of Libraries	-0.0695^{*} ($p = 0.072$)	0.1027^{**} ($p = 0.032$)	0.0047 ($p = 0.953$)
Streetlight Density	1.7757^{**} ($p = 0.038$)	0.0723 ($p = 0.921$)	0.8840 ($p = 0.438$)
No. of Fountains	-0.5208 ($p = 0.106$)	-0.3118 ($p = 0.166$)	-0.3605 ($p = 0.428$)
No. of benches pc	-0.0289 ($p = 0.835$)	-0.5272^{***} ($p = 0.000$)	-0.3600^{**} ($p = 0.031$)

Notes: p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Interpretation and Discussion

Appendix

Declaration of Autorship

I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical

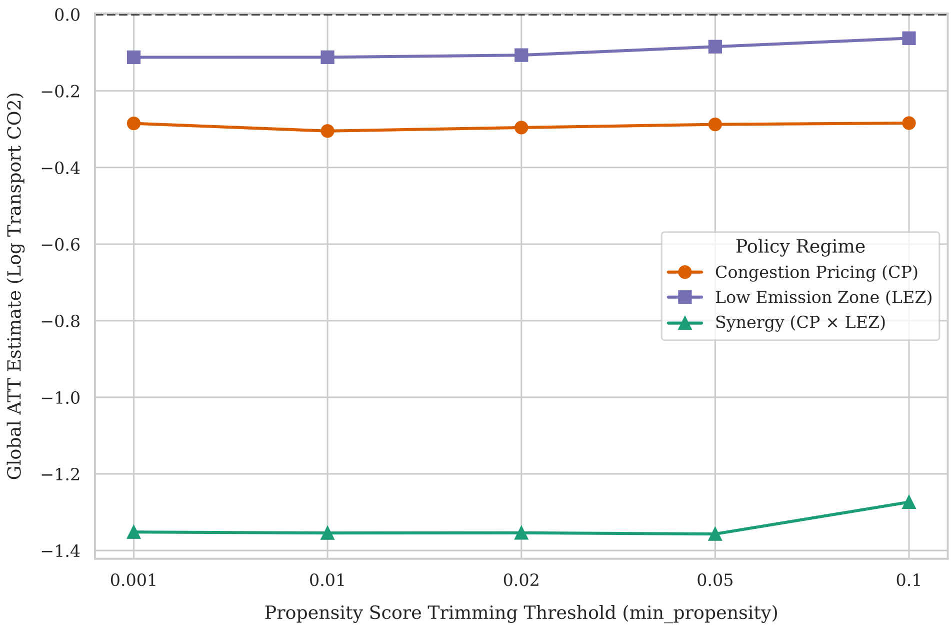

Figure 4: Sensitivity to Positivity Trimming

project with others.

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